Guided Lab: End-to-End Resource Monitoring & Observability in Google Cloud

Overview & Objectives

In this hands-on lab, you will configure end-to-end monitoring, logging, and observability for a Google Cloud VM instance.

By the end of this lab, you will be able to:

1. Deploy a Linux Virtual Machine instance on Compute Engine.
2. Install and verify the Google Cloud Ops Agent for OS-level metrics (RAM, swap, disk space) and system logs across standard and non-LTS Ubuntu releases.
3. Build and test custom Cloud Monitoring Dashboards and Alerting Policies.
4. Configure an automated Log Router Sink to stream operational and audit logs to Cloud Storage for compliance archiving.

Lab Prerequisites

- A Google Cloud account with an active project.
- Basic familiarity with Linux command-line operations.

Task 1: Create a Test Compute Engine VM Instance

1. In the Google Cloud Console, navigate to **Compute Engine** > **VM instances**.
2. Click **Create Instance**.
3. Configure the VM properties using the target parameters:

Property	Value
Name	`demo-monitor-vm`
Region	`us-central1`
Zone	`us-central1-a`

Property	Value
Machine Type	`e2-micro` (or `e2-medium`)
Boot Disk	Ubuntu 24.04 LTS (or Ubuntu 22.04 LTS)

4. Click **Create** and wait for the instance to transition to the running state.

Task 2: Install and Verify the Google Cloud Ops Agent

The built-in hypervisor only collects host-level metrics (e.g., CPU utilization, network traffic). To capture guest OS metrics like memory usage and process logs, install the Google Cloud Ops Agent.

1. In the **VM instances** page, click **SSH** next to `demo-monitor-vm` to open a terminal session.
2. Update the system package repository and install required dependencies:

sudo apt-get update

sudo apt-get install -y curl ca-certificates gnupg

3. Download the official Google Cloud Ops Agent installation script:

curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh

4. Run the installer script based on your distribution release:

- **Standard Installation** (Ubuntu 22.04 / 24.04 LTS):

sudo bash add-google-cloud-ops-agent-repo.sh --also-install

- **Fallback / Non-LTS Releases** (e.g., Ubuntu 26.10 or custom releases):

REPO_CODENAME=noble sudo -E bash add-google-cloud-ops-agent-repo.sh --also-install

5. Verify that the agent service is active and running:

sudo systemctl status google-cloud-ops-agent

Task 3: Explore Memory Metrics in Cloud Monitoring

1. In the Cloud Console, search for **Monitoring**.
2. In the left navigation menu, click **Dashboards**, then select **Metrics Explorer**.
3. Configure the metric selection:
    - **Resource Type:** VM Instance (`gce_instance`)
    - **Metric:** Memory utilization
    - **Filter:** `instance_name = demo-monitor-vm`
4. Confirm that guest memory utilization graphs are rendering in real time.

Task 4: Create a Custom Alerting Policy

1. In the Cloud Monitoring navigation pane, click **Alerting**.
2. Click **Create Policy**.
3. Configure the Metric Condition:
    - **Metric:** VM Instance > CPU utilization
    - **Filter:** `instance_name = demo-monitor-vm`
    - **Condition:** Above threshold
    - **Threshold Value:** `0.80` (80%)
    - **Duration:** 5 minutes
4. Configure Notifications: Select Notification Channels and add your email.
5. Name the Alert: `High CPU Utilization Alert - demo-monitor-vm`
6. Click **Create Policy**.

Task 5: Configure Log Routing to Cloud Storage for Compliance

1. **Create a Destination Storage Bucket:**
    - Go to **Cloud Storage** > **Buckets**
    - Click **Create**
    - **Name:** `audit-logs-archive-demo-unique-id`
    - **Storage Class:** Standard
2. **Create a Log Sink in Log Router:**
    - Go to **Logging** > **Log Router**
    - Click **Create Sink**
    - **Sink Name:** `storage-audit-log-sink`

- **Sink Destination:** Cloud Storage bucket
- **Select Bucket:** `audit-logs-archive-demo-unique-id`
- **Inclusion Filter:** `severity >= WARNING`

3. Click **Create Sink**.

Task 6: Verify Log Archival and Clean Up

1. Return to your SSH session on `demo-monitor-vm` and generate a sample log:

```
logger -p syslog.warn "Testing Google Cloud Logging Sink Routing"
```

2. Navigate to **Cloud Storage** > **Buckets** and open your archive bucket to confirm log directories are created.

Lab Cleanup

To prevent incurring unnecessary charges, clean up the resources created during this lab:

1. Delete the VM instance `demo-monitor-vm`.
2. Delete the alerting policy `High CPU Utilization Alert - demo-monitor-vm`.
3. Delete the log sink `storage-audit-log-sink`.
4. Delete the Cloud Storage bucket `audit-logs-archive-demo-unique-id`.